

Yu, Zijun

Email: zijun.yu@mail.mcgill.ca

Education

- 2025 – Present** **Ph.D. in Mathematics and Statistics**
McGill University, Montreal, QC
Research Interest: Neural Networks, Survival Analysis
- 2016 – 2020** **B.Sc. Honours in Statistics and Computer Science**
McGill University, Montreal, QC
First-Class Honours Degree

Awards and Scholarships

- Graduate Excellence Award 2025-
- ISM Graduate Scholarship 2025-2026
- First-Class Honours Degree 2020
- Science Undergraduate Research Award (SURA) 2019
- James McGill Scholarship 2016

Publications and Presentations

- Pengqi Liu[†], **Zijun Yu**[†], Mouloud Belbahri, Arthur Charpentier, Masoud Asgharian, Jesse C. Cresswell. “Beyond Procedure: Substantive Fairness in Conformal Prediction.” *International Conference on Machine Learning (ICML)*, 2026. [arXiv:2602.16794](https://arxiv.org/abs/2602.16794)
- Yu Gu[†], **Zijun Yu**[†], Vahid Partovi Nia, Masoud Asgharian. “Pause and Reflect: Conformal Aggregation for Chain-of-Thought Reasoning.” *Under review*, 2026.
- Yu Gu[†], **Zijun Yu**[†], Chi-Kuang Yeh, Xinyu Wang, Ziyang Song. “Severity-Controlled Prediction Sets for Medication Recommendation.” *Under review*, 2026.

[†] Equal contribution.

Work Experience

2025 – 2026 *Mitacs Internship — Student Researcher*
Layer6 AI, Toronto

- Conducted research on substantive fairness in conformal prediction, contributing to the paper “Beyond Procedure: Substantive Fairness in Conformal Prediction” (ICML 2026).
- Designed and architected a modular codebase to support large-scale experiment pipelines, enabling reproducible benchmarking across multiple fairness criteria, datasets, and model families.
- Developed novel post-hoc prediction set methods that incorporate group fairness constraints without sacrificing marginal coverage guarantees.
- Collaborated with research scientists on theoretical analysis and empirical validation under the supervision of Jesse C. Cresswell and Mouloud Belbahri.

2020 – 2024

Software Developer

OneDesk Inc., Montreal

- Incorporated data analysis and visualization capabilities into existing software solutions
- Implemented a Retrieval-Augmented Generation (RAG) chatbot workflow with web crawler, vector database, and OpenAI APIs.
- Developed third-party integrations and plugins for platforms such as Microsoft Teams App, Google OAuth, WordPress Plugin, Stripe payment processing, and Intuit invoice processing.
- Applied time-series analysis and correlation to analyze system logs, monitor system performance, and identify trends.
- Redesigned and refactored both frontend and backend systems to fix memory leaks and resolve data inconsistencies, ensuring robustness under race conditions and optimizing performance.
- Led a team of junior developers, providing mentorship and supervision across multiple projects. Coordinated team efforts to meet project goals efficiently.

Technical Skills

Python, R, Java (Spring, Maven, GraalVM), TypeScript (React, Deno), Redis

SQL (PostgreSQL), Virtualization (Docker, Proxmox VE), Version Control System (Git, Subversion)